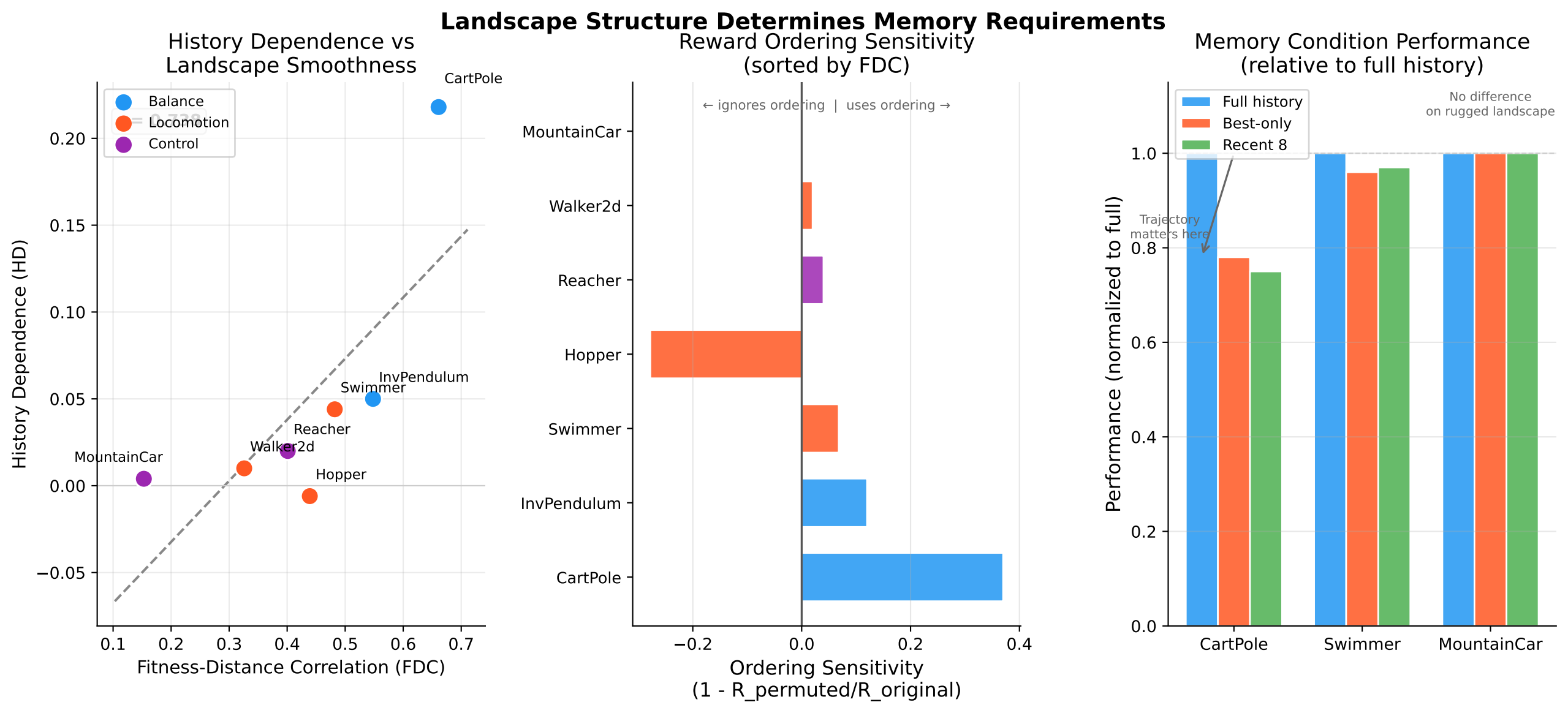
Landscape Structure Determines Memory Requirements
History Dependence vs
Landscape Smoothness
Balance
Locomotion
Control
CartPole
InvPendulum
Swimmer
Reacher
Walker2d
MountainCar
Hopper
History Dependence (HD)
Fitness-Distance Correlation (FDC)
Reward Ordering Sensitivity
(sorted by FDC)
← ignores ordering | uses ordering →
MountainCar
Walker2d
Reacher
Hopper
Swimmer
InvPendulum
CartPole
Ordering Sensitivity
(1 - R_permuted/R_original)
Memory Condition Performance
(relative to full history)
Full history
Best-only
Recent 8
No difference
on rugged landscape
Trajectory
matters here
Performance (normalized to full)
CartPole
Swimmer
MountainCar